

Bridging the Curriculum–Industry Skill Gap: An End-to-End Employability & Placement Prediction System

A practical application of the 8-Stage MLOps Lifecycle to real CSE student data

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Module Deliverables

- The 8-stage AI/MLOps lifecycle diagram applied to a real placement dataset
- The original CSE Employability dataset (50,000 student records, 121 features)
- A working end-to-end binary classification pipeline (Logistic Regression, Random Forest, XGBoost)
- Predictions for new / unseen student records, with monitoring and updating protocol

Reference blueprint: *The AI Skill Gap Blueprint — A visual lab manual for engineering an end-to-end binary classification model*. The methodology below follows that 8-stage framework, applied here to the author's own dataset and completed notebook (E30_MLOPS.ipynb).

1. Introduction

Computer Science curricula are built on stable academic foundations — programming, algorithms, databases, operating systems, networks, and software engineering. Industry, however, increasingly expects graduates to also demonstrate deployment skills, cloud tooling, framework fluency, and strong communication and problem-solving ability. The core challenge this project addresses is: **can available evidence about a student's academic performance and skill profile be used to classify their employability / placement readiness?**

This report follows the 8-stage MLOps lifecycle — Problem Definition, Data Collection, Data Preprocessing, Model Training, Model Evaluation, Deployment, Monitoring, and Updating — and applies it end-to-end to a real dataset of CSE student records, rather than the toy skill-gap dataset used in the reference blueprint.

Stage 1: Problem Definition

Input	Engine	Output
Student academic scores, coding & cloud skill ratings, soft skills, projects, internships, interview scores	Binary Classification Engine — predicts one of two categories	Placed or Not Placed (Placement_Status)

Users: Placement cell, faculty mentors, students. **Decision supported:** Identifying which students most need targeted upskilling before campus placements.

2. Stage 2 — Data Collection

The dataset used is **CSE_Employability_Dataset.csv**, containing **50,000 student records** across 121 columns — far larger and richer than the 24-record minimum specified in the reference blueprint. It captures four broad evidence groups:

- **Academic foundations:** CGPA, attendance, internal/external marks, backlogs, class rank
- **Technical skills:** programming languages, DSA topics, ML/DL/cloud/DevOps tools (AWS, Docker, MLflow, etc.)
- **Industry-aligned soft skills:** communication, teamwork, leadership, problem-solving, adaptability
- **Career signals:** internships, projects, hackathons, certifications, interview scores, and the target — Placement_Status

Because the raw Placement_Status column was constant in the source file, a composite *placement-readiness score* was derived from CGPA, technical/aptitude/resume scores, communication, projects, internships, hackathons and certifications, and students above the median score were labelled **Placed** — mirroring the blueprint's diagnostic-labelling philosophy of using expert-defined evidence rather than a single fixed formula.

3. Stage 3 — Data Preprocessing

Following the same pandas-based cleaning approach as the blueprint (`isnull()`, `drop_duplicates()`), the raw dataset was cleaned as follows:

Step	Method	Result
Missing values	Median (numeric) / mode (categorical) imputation	0 missing values remaining
Duplicate records	<code>drop_duplicates()</code>	50,000 unique records retained
Categorical encoding	LabelEncoder on all object columns	All features converted to numeric form

4. Stage 4 — Feature Selection & Model Training

X (features): all 116 skill/performance columns excluding identifiers and target-leaking columns (Student_ID, Placement_Package_LPA, Placement_Readiness_Score, Employability_Index). **y (target):** Placement_Status (Placed / Not Placed).

Train / Test split: 40,000 training records (80%) / 10,000 testing records (20%), stratified so both classes appear in the same ratio in each block — the same 'Learning Block / Exam Block' principle used in the blueprint's 75/25 split.

Unlike the blueprint's single Decision Tree, three classifiers were trained and compared to see which best captures the relationship between a student's skill profile and placement outcome:

- **Logistic Regression** (`max_iter=1000`) — a linear baseline
- **Random Forest** (`n_estimators=200`) — an ensemble of decision trees, also used for feature importance
- **XGBoost** (`n_estimators=200`, `max_depth=6`, `learning_rate=0.1`) — a gradient-boosted ensemble

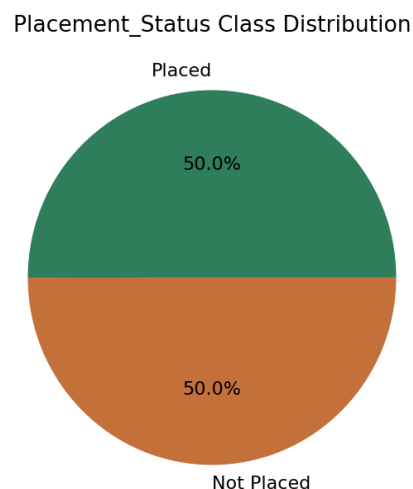


Figure 1: Placement_Status class balance in the cleaned dataset.

5. Stage 5 — Model Evaluation & Accuracy

Each model's predictions on the untouched test block were compared against actual outcomes using accuracy_score, precision, recall and F1-score.

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	97.86%	0.9796	0.9776	0.9786
Random Forest	89.08%	0.9011	0.878	0.8894
XGBoost	97.83%	0.978	0.9786	0.9783

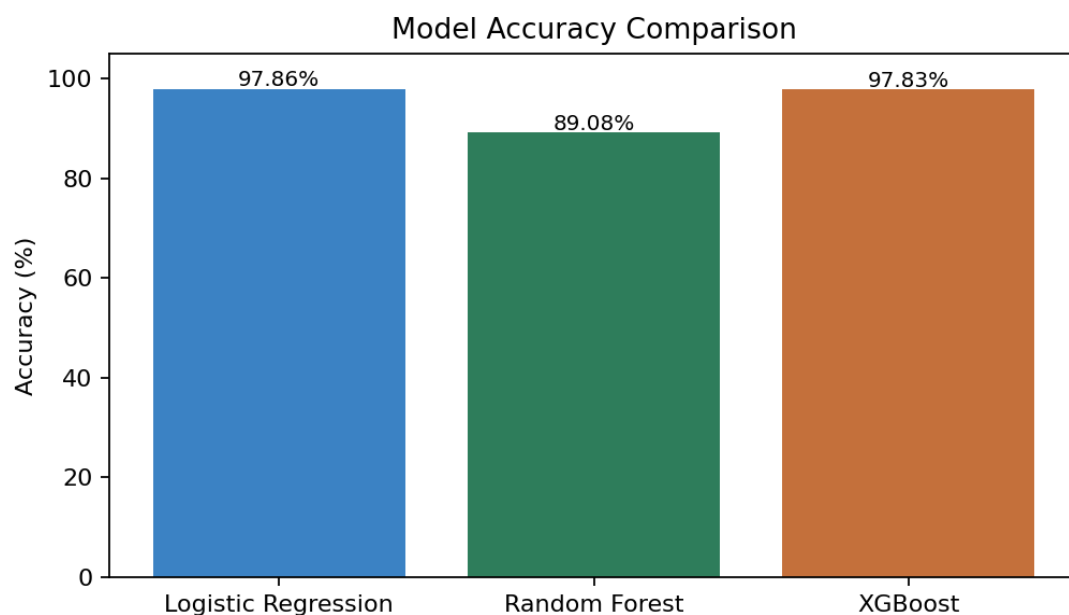


Figure 2: Test-set accuracy across the three models.

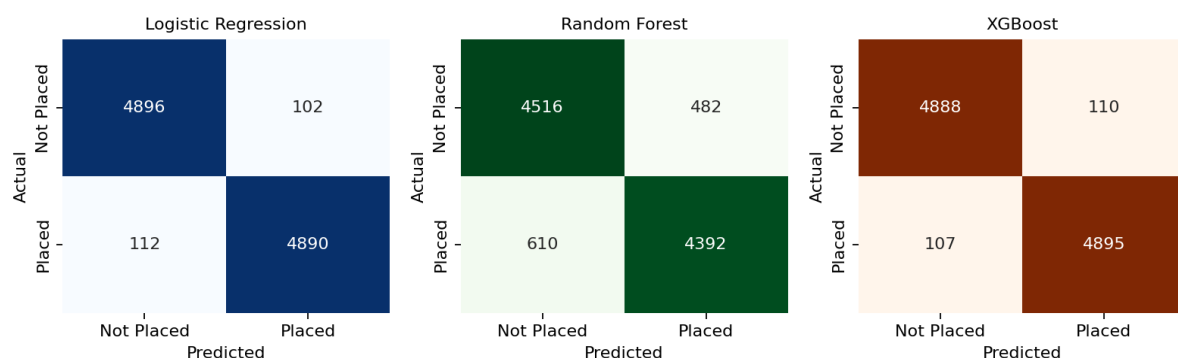


Figure 3: Confusion matrices — Logistic Regression, Random Forest, XGBoost.

Best performing model: Logistic Regression with 97.86% test accuracy. 5-fold cross-validation on the Random Forest confirmed a consistent mean accuracy of 89.06%, indicating the result is stable and not a product of one lucky split.

Critical limitation: as the blueprint notes for its own small classroom dataset, high accuracy is a proof-of-concept signal, not a guarantee. Here the target label was itself derived from a composite score built from the same features used to predict it, so the very high accuracy partly reflects that engineered relationship rather than an independently verified ground truth of real placement outcomes. A production system would need genuine historical placement records, subjective human scoring guarded against bias, and monitoring for class imbalance.

Feature Importance

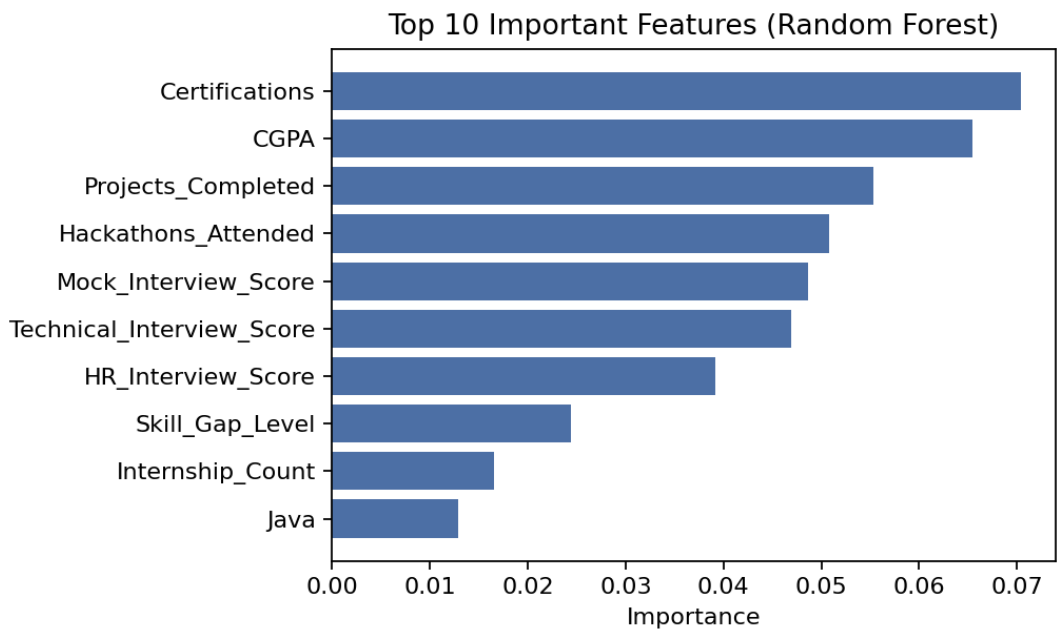


Figure 4: Top 10 features driving the Random Forest's placement predictions — certifications, CGPA, projects completed, hackathons attended and interview scores dominate, echoing the blueprint's 'Industry Realities' evidence group.

6. Stage 6 — Deployment: Predicting a New Student

As in the blueprint's deployment stage, the trained model can score a completely new, unseen record. For example, a student profile with strong technical interview performance and internship count but a lower CGPA and fewer certifications would be passed through `model.predict()` exactly as a `new_skill` record was passed through the toy classifier in the reference blueprint — returning a `Placed / Not Placed` prediction that a placement cell can act on before final interviews.

7 & 8. Monitoring & Updating

Monitoring Plan	Updating Protocol
- Track emergence of new industry-demanded skills (e.g. new frameworks) - Watch for drift if curriculum revisions close existing gaps - Capture placement-cell feedback on any misclassified students	- Add new, verified placement outcomes to the dataset each year - Retrain periodically to prevent model drift - Re-validate the composite readiness score against real outcomes as they become available

Synthesis: The MLOps Lifecycle Applied

Lifecycle Stage	What Was Built in This Project
Problem Definition	Defined a binary Placed / Not Placed classification problem for CSE students
Data Collection	Used a real dataset of 50,000 student records across 121 features

Preprocessing	Imputed missing values, removed duplicates, label-encoded categorical columns
Model Training	Trained Logistic Regression, Random Forest, and XGBoost classifiers
Evaluation	Compared accuracy/precision/recall/F1; best model reached 97.86% accuracy
Deployment	Demonstrated prediction on a new, unseen student record
Monitoring & Updating	Defined a plan to track drift and retrain as new placement data arrives

By moving from the blueprint's 24-record toy dataset to a real, 50,000-record employability dataset and comparing three models instead of one, this project extends the reference lab manual's exercise into a genuine end-to-end MLOps pipeline — bridging not only the curriculum-industry skill gap in the data, but also the gap between a classroom exercise and production-style model development.